

CS-302: Artificial Intelligence

LECTURE NO 2:

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Organization of Lecture

1. Types of Agents
 1. Simple Reflex Agents
 2. Model-based Reflex Agents
 3. Goal-based Agents
 4. Utility-based Agents
 5. Learning Agents
2. PEAS: Group of AI Agents

1. Types of Agents (1/7)

Types of Agents

→ AI is defined as study of Rational Agents

→ AI Agent is responsible for any work output obtained from the system.

Agent
 → Information Gathering
 → Based on information gathering it can perform some actions.

→ Rational AI Agents
 → Person
 → Firm
 → Machine/S/w } that can make decisions

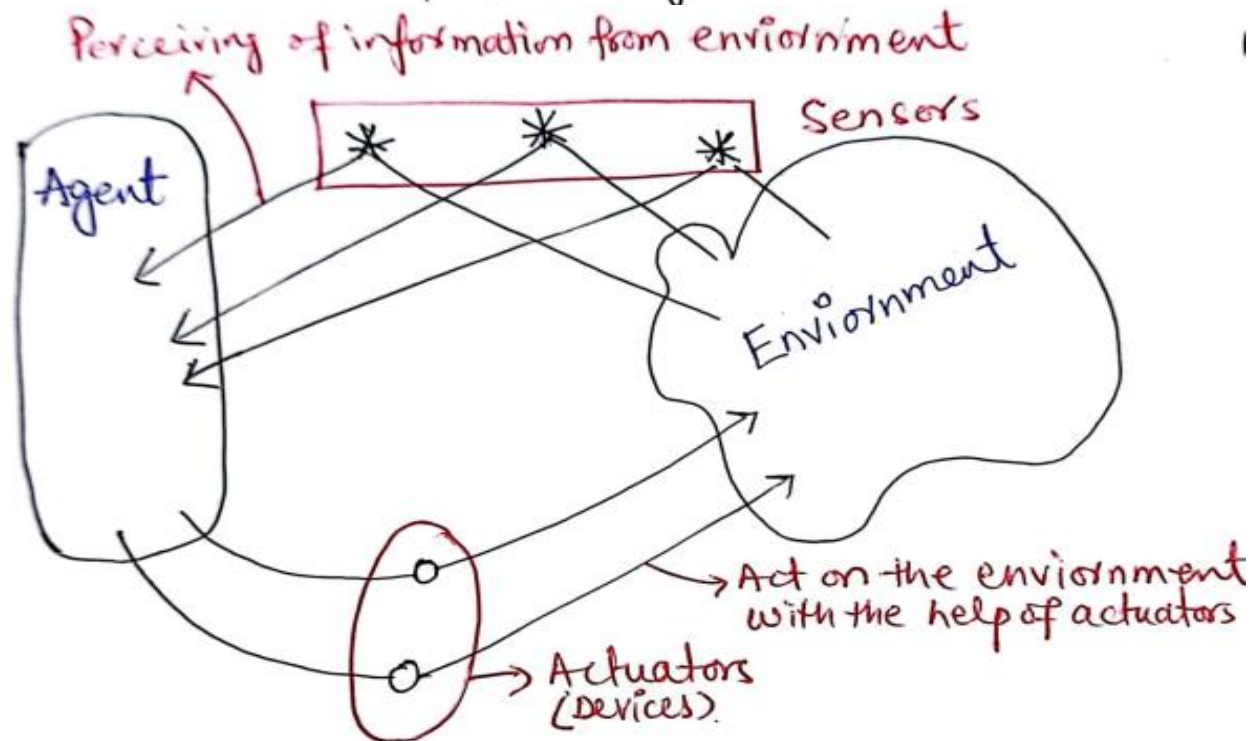
→ AI system is composed of
 → Agent
 → Environment

1. Types of Agents (2/7)

→ Agent is anything like:

- i) perceiving its environment through **Sensors**.
- ii) Acting upon that environment through **actuators**.

→ Let's understand it with the help of a diagram.



1. Types of Agents (3/7)

① Simple Reflex Agents

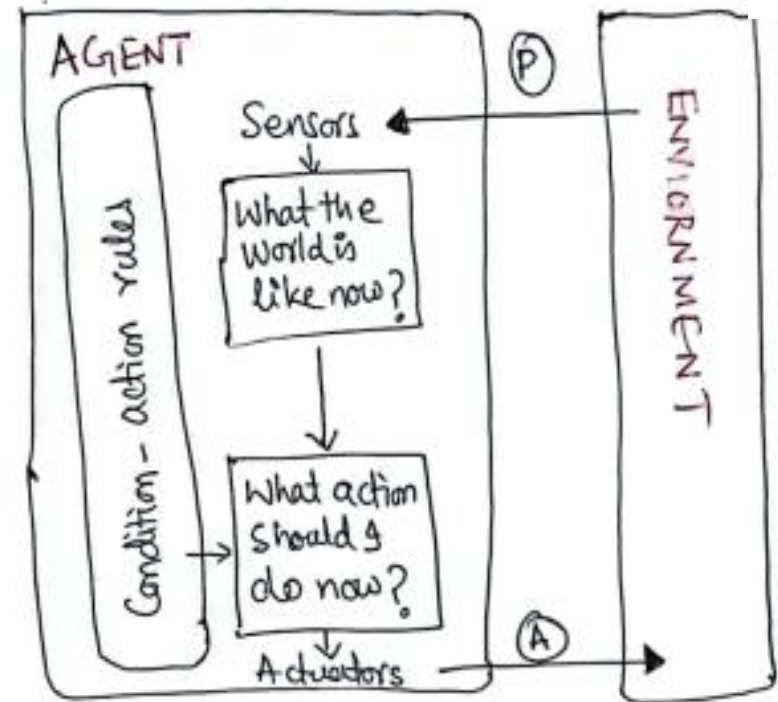
→ Works only on Current Situation/perception and ignores the history of previous state.

→ Condition-Action Rule

- If condition is TRUE, then take some action.
- Condition-Action Rules are already pre-defined.

Limitations:

- Very Limited Knowledge or Intelligence
- No knowledge about non-perceptual parts of state.
- Operating in a partially observable environment can cause infinite loops.



(P) = Perception

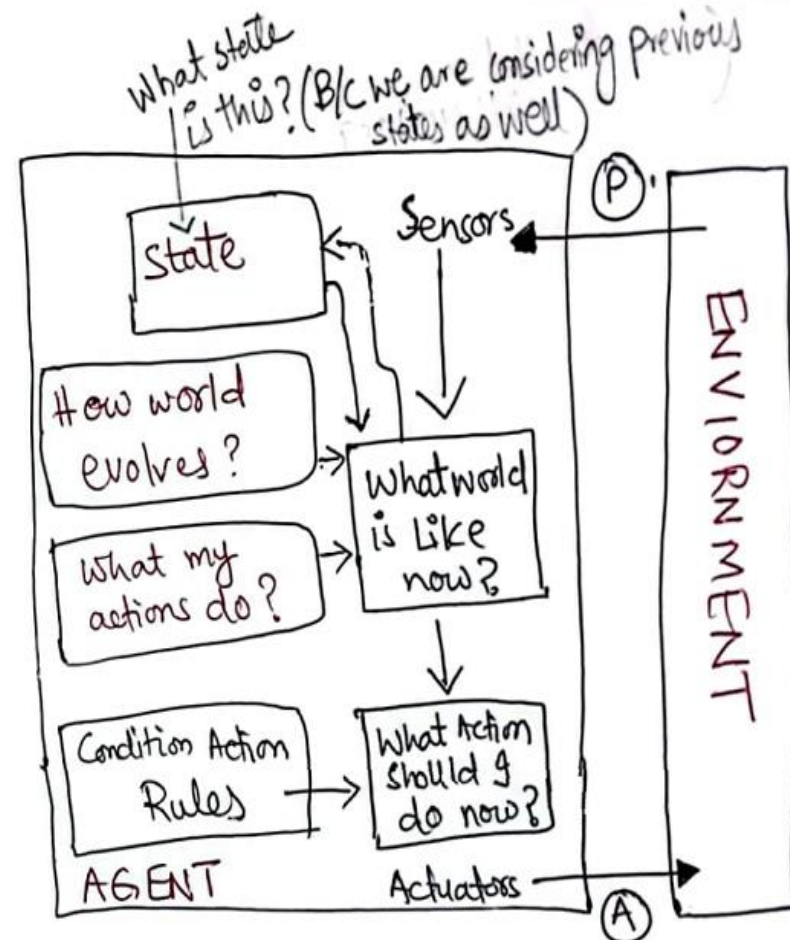
(A) = Action

1. Types of Agents (4/7)

② Model based Reflex Agent:

- ↳ Works by finding the rule whose condition matches current situation.
- ↳ Can work in partially observable environment, and track situation.
- ↳ Agent keeps track of internal state which is adjusted by each percept and that depends on percept history: (maintaining previous history/information).
- ↳ Model: How things happen in world?

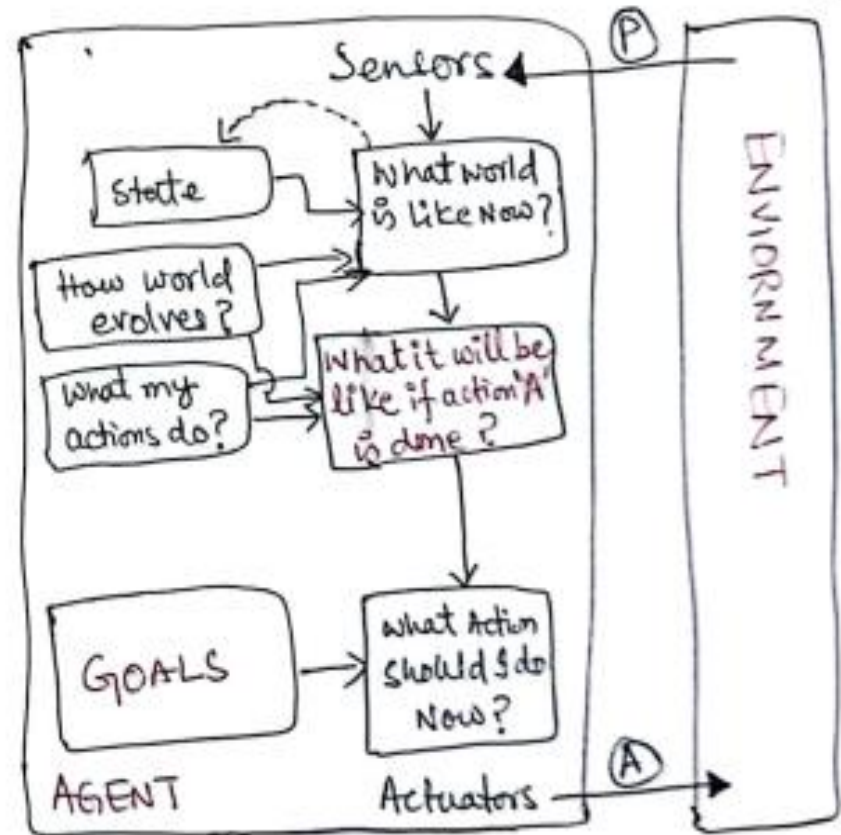
- ↳ Agent state updation required
Information:
 - ① How world is evolving?
 - ② How an agent's action affects the world?



1. Types of Agents (5/7)

③ Goal-Based Agents

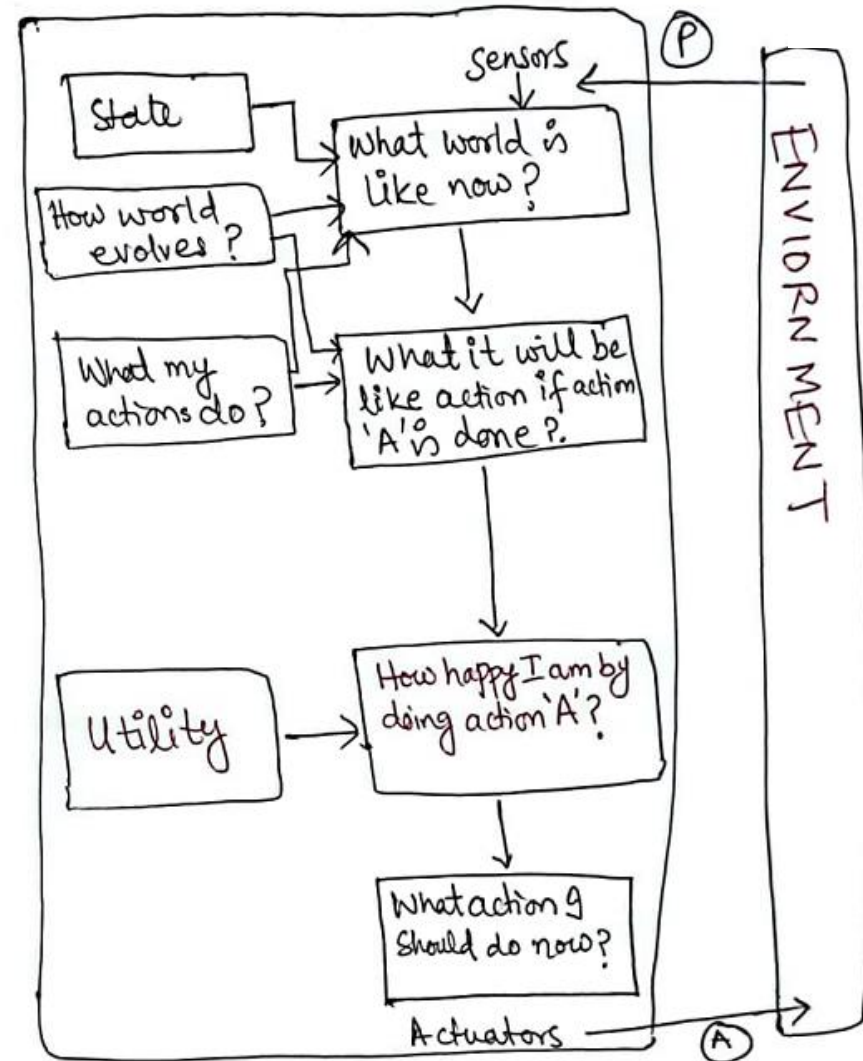
- ↳ Focuses only on reaching the goal set.
- ↳ Agent takes decision based on how far it is currently from the Goal State.
- ↳ Every action is taken to minimize distance to Goal state.
- ↳ More flexible Agent.



1. Types of Agents (6/7)

④ Utility-Based Agents

- Agents are more concerned about the utility (Preference) for each state. (Preference of each state is considered).
- Act based not only on goals but also the best way to achieve goal.
- Useful when there are multiple possible alternatives and agent has to choose in order to perform best action.



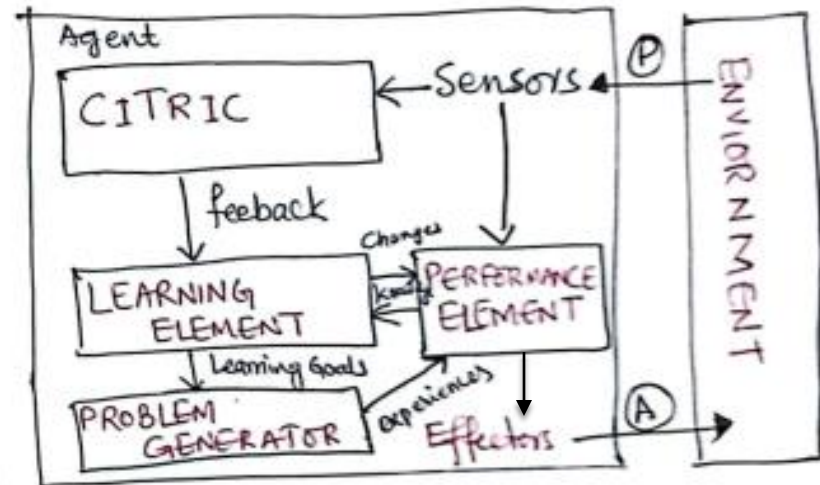
1. Types of Agents (7/7)

⑤ Learning Agents

- ↳ can learn from its past experiences.
- ↳ Starts to ACT with basic knowledge and then able to act by adapting learning.

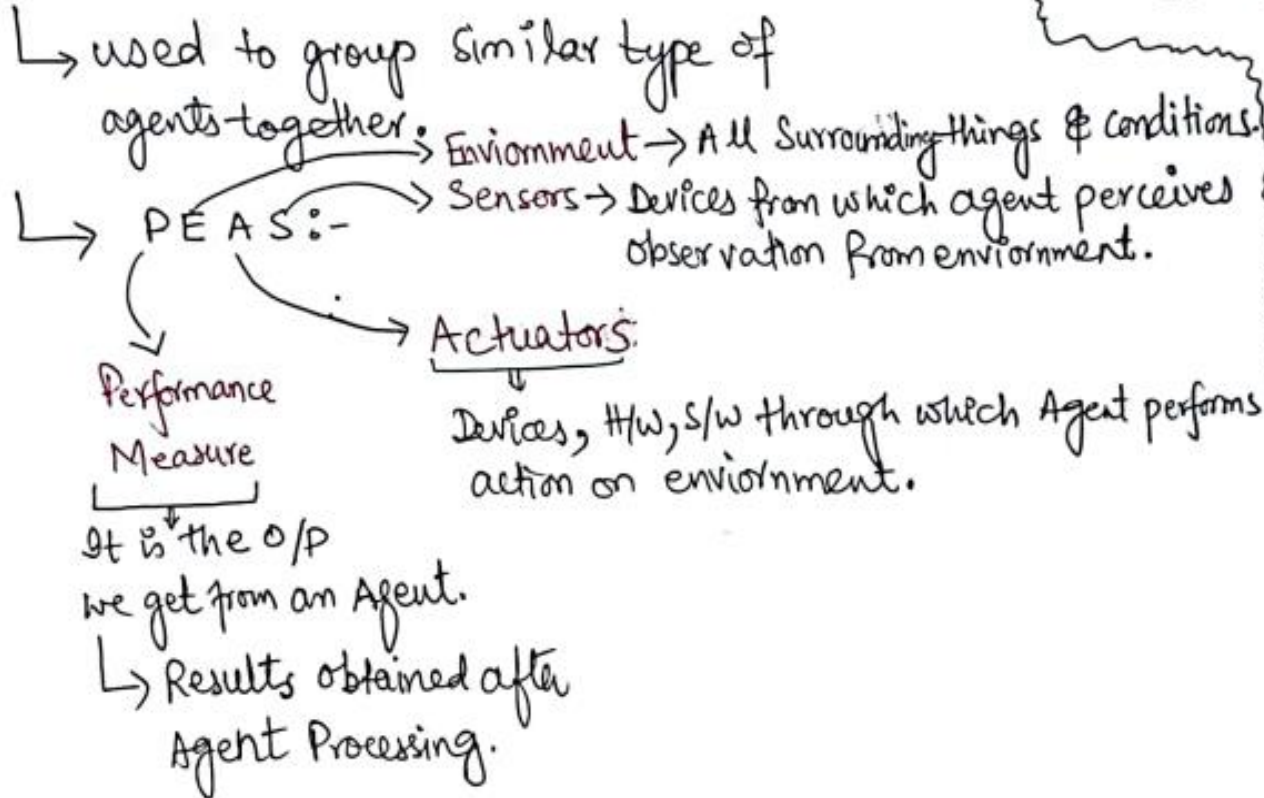
Components

- Learning Element: Makes improvement in the system by learning from environment.
- Citric: Gives feedback about agent's performance based on certain standard.
- Performance Element: Selects the action to Perform
- Problem Generator: Suggests the action → for gaining new information.



2. PEAS: Group of AI Agents

PEAS: Grouping of AI Agents:



Example:-

SELF DRIVING CAR

Performance Measure (P)

→ Comfort, Safety, Time, legal driving.

Environment (E)

→ Condition of Roads, crossings, Traffic Signals.

Actuators (A)

→ Steering, Brakes, Horn, Accelerator.

Sensors (S)

→ Camera, GPS, Speedometer, etc.

References: Textbooks

1. *“Artificial Intelligence”, by Elaine Rich and Kevin Knight, (2006), McGraw Hill companies Inc., Chapter 1-22 page 1-613.*
2. *“Artificial Intelligence: A Modern Approach” by Stuart Russell and Peter Norvig, (2009), Prentice Hall, Chapter 1-27. page 1-1151.*
3. *“Computational Intelligence: A logical Approach”, by David Poole, Alan Mackworth, and Randy Goebel, (1998), Oxford University Press, Chapter 1-12, page 1-608.*
4. *“Artificial Intelligence: Structures and Strategies for Complex Problem Solving”, by George F.Lugar, (2002), Addison-Wesley, Chapter 1-16, page 1-743.*
5. *“AI: A new Synthesis”, by Nils J.Nilsson, (1998), Morgan Kaufmann Inc., Chapter 1-25, Page 1-493.*
6. *“Artificial Intelligence: Theory and Practice” by Thomas Dean, (1994), Addison-Wesley, Chapter 1-10, Page 1-650.*
7. *Related documents from open source, mainly internet.*